Problem 10 – ProofCouncil

Mirror fibrations for very general H-polarised Tyurin degenerations of K3 surfaces

1 Introduction

1.1 The statement

Throughout, H is the hyperbolic plane lattice (even, unimodular, of signature $(1,1)$), E_8 the negative definite E_8 root lattice, $\Lambda_{\mathrm{K3}} = H^{\oplus 3} \oplus E_8^{\oplus 2}$ the K3 lattice, and $M = H \oplus E_8 \oplus E_8$. For an even lattice N of signature $(1, \operatorname{rk} N - 1)$, an *$N$-quasipolarised* K3 surface is a K3 surface Y with a primitive embedding $N \hookrightarrow \mathrm{NS}(Y)$ satisfying the chamber convention of [GT24, Definition 3.19] (the embedding sends a fixed very irrational positive vector of $N \otimes \mathbb{R}$ to a nef and big class; the convention goes back to [Do96]); in particular the image of N contains a nef and big class. For very general such Y (outside countably many Noether–Lefschetz divisors of the moduli space) $N \simeq \mathrm{NS}(Y)$. In Dolgachev's mirror symmetry for lattice-polarised K3 surfaces [Do96] the lattices H and M are mirror to each other, as $H^{\perp}_{\Lambda_{\mathrm{K3}}} \simeq H \oplus M$.

A *Kulikov model* is a proper flat family $\pi\colon \mathcal{X} \to \Delta$ over a disc with smooth total space, trivial canonical bundle, reduced normal crossings central fibre and smooth K3 general fibre [Ku77]. A *Tyurin degeneration* is a Kulikov model whose central fibre $X_0 = V_1 \cup V_2$ is the union of two smooth rational surfaces meeting transversally along a smooth elliptic curve C (a Kulikov model of *Type II*: the double curves are elliptic). By adjunction C is anticanonical on each V_i, and smoothness of $\mathcal{X}$ makes X_0 *d*-semistable [Fr83]: $N_{C/V_1} \otimes N_{C/V_2} \simeq \mathcal{O}_C$, so $(C^2)_{V_1} + (C^2)_{V_2} = 0$.

Following the proposal of Doran, Harder and Thompson [DHT17] that Tyurin degenerations should be mirror to fibrations over $\mathbb{P}^1$, Giovenzana and Thompson [GT24] define when an N-polarised Tyurin degeneration (Definition 1.1 below, for $N = H$) and an elliptic fibration on the mirror K3 surface, split into two pieces by a loop in $\mathbb{P}^1$, form a *mirror pair* [GT24, Definition 5.1]; the relevant notions are recalled in §2.

Definition 1.1. A Tyurin degeneration $\pi\colon \mathcal{X} \to \Delta$ is *H-polarised* if (1) (monodromy invariance) over the punctured disc Δ^* there is a trivial local subsystem $\mathcal{L} \subset R^2\pi_*\mathbb{Z}$ with fibre H, such that for each $t \in \Delta^*$ the fibre $\mathcal{L}_t \subset H^2(X_t;\mathbb{Z})$ defines an H-quasipolarisation of X_t; and (2) (positivity) there is an effective, flat (over Δ, i.e. with no component inside X_0) and

nef Cartier divisor A on $\mathcal{X}$ whose restriction A_t to every fibre X_t, $t \in \Delta^*$, is nef and big with $[A_t] \in \mathcal{L}_t$, and whose restriction to each component satisfies $(A|_{V_i})^2 > 0$, $i = 1, 2$.

This is [GT24, Definition 3.20] with two strengthenings of (2) that are part of our hypotheses: the membership $[A_t] \in \mathcal{L}_t$, which corrects the arXiv version of that definition, and the componentwise positivity $(A|_{V_i})^2 > 0$.

Theorem 1.2. *Let $\pi\colon \mathcal{X} \to \Delta$ be an H-polarised Tyurin degeneration of K3 surfaces with central fibre $X_0 = V_1 \cup V_2$. Assume it is very general, in the sense that the image of $0 \in \Delta$ under the map from Δ to the Alexeev–Brunyate–Engel compactification [ABE22] of the moduli space of H-polarised K3 surfaces lies in no Noether–Lefschetz locus. Let Y be a very general M-quasipolarised K3 surface. Then there exist an elliptic fibration $Y \to \mathbb{P}^1$ and an allowable loop $\gamma \subset \mathbb{P}^1$ [GT24, Definition 4.11] such that the M-quasipolarisation is compatible with the fibration and γ [GT24, Definition 4.21], and such that $\mathcal{X} \to \Delta$ and $(Y \to \mathbb{P}^1, \gamma)$ form a mirror pair in the sense of [GT24, Definition 5.1].*

1.2 Conventions and interpretation

We use the definitions of [GT24] in arXiv version 3. Noether–Lefschetz loci on the compactification are the closures of the Noether–Lefschetz loci of its interior. All lattices and homology groups are integral; $x \cdot y$ or (x, y) denotes the bilinear form, $x^2 = (x, x)$, and a *root* is a vector of square -2.

The degeneration determines more than the image point of 0 in the ABE compactification: it determines the *marked* ("mixed-marked" in [AE23]) Type II period of its central fibre, namely the isotropic plane I of the monodromy (§2.2) and the restriction homomorphism φ of §3. We always work with this marked period. The distinction matters: over one of the two Type II boundary strata the boundary normalisation map of [ABE22, Theorem 7.58] has degree eight, so a coarse boundary point does not determine a marked period; we never pass from one to the other, and no base change or level cover is introduced.

1.3 Strategy

Both sides produce an even unimodular lattice of signature $(1, 17)$ with a distinguished sublattice and two mutually orthogonal sublattices (one for each component of X_0, respectively each piece of the split fibration); a mirror pair is essentially an isometry of these rank-18 lattices matching the embedded sublattices and lifting to the K3 lattices (compare [GT24, Proposition 5.3]). On the degeneration side the hypotheses do not describe X_0 explicitly. Section 3 shows that very generality forces the kernel of the restriction homomorphism to be the polarising lattice $L \simeq H$, so the rank-18 lattice is $L \oplus R_X$ with $R_X \simeq E_8^2$ or D_{16}^+ (notation of §2). Section 4 constructs, for the given marked period, an explicit Kulikov model $\mathcal{W}$ isomorphic to $\mathcal{X}$ over Δ^* and in codimension one. Since Kulikov models are unique only up to elementary modifications (flops along curves in the central fibre), $\mathcal{W}$ need not be $\mathcal{X}$; Section 5 transports the divisor A to $\mathcal{W}$ and compares the contractions of $\mathcal{W}$ and $\mathcal{X}$ defined by A. Componentwise positivity of A leaves exactly three central fibres, $\mathrm{E}_0, \mathrm{E}_1$ (for $R_X = E_8^2$)

and D (for $R_X = D_{16}^+$), whose embedded lattices are computed in §6. On the fibration side, §7 exhibits two elliptic fibrations on Y (the *standard* one, with singular fibres $2\,\mathrm{II}^* + 4\,\mathrm{I}_1$, and the *alternate* one, with $\mathrm{I}_{12}^* + 6\,\mathrm{I}_1$) and three allowable splitting loops $\gamma_0, \gamma_1, \gamma_D$; §8 proves compatibility and computes the embedded lattices, which match the degeneration cases according to $\mathrm{E}_0 \leftrightarrow (\text{standard}, \gamma_0)$, $\mathrm{E}_1 \leftrightarrow (\text{standard}, \gamma_1)$, $\mathrm{D} \leftrightarrow (\text{alternate}, \gamma_D)$. The delicate points are integral: the small piece of γ_D has "degree $8'$" rather than 8, and the large piece carries an index-two parity condition (§8). Section 9 lifts the isometry to the K3 lattices and checks every clause of [GT24, Definition 5.1].

2 Notation and recollections

2.1 Lattices

D_{16} is the negative definite D_{16} root lattice and D_{16}^+ its even unimodular index-two overlattice; E_8^2 and D_{16}^+ are the only even unimodular negative definite lattices of rank 16 [Se62, §8]. $\mathrm{I}_{1,n}$ is the odd unimodular lattice of signature $(1,n)$ and $\langle m \rangle$ the rank-one lattice with generator of square m. A sublattice $S \subset \Lambda$ is *primitive* if Λ/S is torsion free; a vector k is *characteristic* if $k \cdot x \equiv x^2 \pmod 2$ for all x. Two facts from [Ni80] are used repeatedly: a unimodular sublattice $S \subset \Lambda$ is a direct summand, $\Lambda = S \oplus S^\perp$; and if v is a primitive vector of a unimodular lattice with $v^2 \neq 0$, then $v^\perp$ has cyclic discriminant group of order $|v^2|$.

2.2 The degeneration side

Let $\pi\colon \mathcal{X} \to \Delta$ be as in Theorem 1.2, with $C = V_1 \cap V_2$ and $c_i = [C] = -K_{V_i} \in \mathrm{NS}(V_i)$. Put

$$\xi = (c_1, -c_2) \in \mathrm{NS}(V_1) \oplus \mathrm{NS}(V_2), \qquad \Lambda_X = \xi^\perp/\mathbb{Z}\xi = I^\perp/I. \tag{1}$$

Here $I \subset H^2(X_t;\mathbb{Z})$ is the primitive isotropic plane of the degeneration (the image of $\log T$, where T is the unipotent monodromy acting on $H^2(X_t;\mathbb{Z})$), and the second equality in (1) is an *integral* isomorphism [GT24, Proposition 3.5]; Λ_X is even, unimodular and of signature $(1,17)$. The monodromy invariant lattice $\mathcal{L}_t \simeq H$ lies in $I^\perp$ and meets I trivially; we write $L \subset \Lambda_X$ for its image, and

$$R_X = L_{\Lambda_X}^\perp, \qquad S_{X,i} = \mathrm{im}\big(c_i^\perp \longrightarrow \Lambda_X\big), \quad \text{where } c_i^\perp \subset \mathrm{NS}(V_i). \tag{2}$$

The maps $c_i^\perp \to \Lambda_X$, $D \mapsto [(D,0)]$ or $[(0,D)]$, are injective, and $S_{X,1} \perp S_{X,2}$ since $(D_1,0)\cdot(0,D_2) = 0$.

2.3 The fibration side

We paraphrase the framework of [GT24, §§2,4,5], which builds on the pseudolattice formalism of [Kuz17, HT20]; the precise definitions are those of [GT24]. Let $Y \to \mathbb{P}^1$ be an elliptic K3 surface with a section and $\gamma \subset \mathbb{P}^1$ a loop avoiding the critical values, with $\mathbb{P}^1 = \Delta_1 \cup_\gamma \Delta_2$

a union of closed discs, Y_i the preimage of Δ_i and F_p the fibre over a base point $p \in \gamma$. The relative homology $H_2(Y_i, F_p)$ is free of rank $e(Y_i)$, the Euler number of Y_i (the sum over the singular fibres in Δ_i), generated, when these fibres are nodal, by Lefschetz thimbles (the discs swept out by the vanishing cycles along paths from p to the critical values). It carries the unimodular *Seifert form* χ_i (upper triangular with ones on the diagonal in a basis of distinguished thimbles, i.e. over an ordered system of disjoint paths, the entries above the diagonal being intersection numbers of vanishing cycles). The boundary map $\phi_i\colon H_2(Y_i, F_p) \to H_1(F_p)$ is the *charge map*, and its *right adjoint* $r_i\colon H_1(F_p) \to H_2(Y_i, F_p)$ is defined by $\chi_i(v, r_i(x)) = \langle \phi_i(v), x\rangle$, where $\langle\ ,\ \rangle$ is the intersection form on $H_1(F_p)$. Fix a symplectic basis a, b of $H_1(F_p)$ with $\langle a, b\rangle = -1$ (the sign convention of [GT24]). The loop is *allowable* [GT24, Definition 4.11] if both pieces satisfy [GT24, Assumption 4.9]: the monodromy around γ, traversed as the anticlockwise boundary of Δ_i, is $\left(\begin{smallmatrix} 1 & e(Y_i)-12 \\ 0 & 1 \end{smallmatrix}\right)$ in the basis (a, b), and $r_i(a)$ is primitive. For such a piece $r_i(a)$ plays the role of the point class of a surface-like pseudolattice [GT24, §§2,4], whose Néron–Severi lattice is $N_i = r_i(a)^\perp/\mathbb{Z}\, r_i(a)$, where $r_i(a)^\perp = \{v : \chi_i(v, r_i(a)) = 0\}$, with the surface-like form $q_i = -\chi_i$ (symmetric there), and whose canonical class is, up to a sign that does not affect its orthogonal complement, $k_i = [r_i(b)] \in N_i$. Since a is monodromy invariant, transporting it around γ sweeps out a torus in Y; let F be the fibre class and $\bar\tau \in F^\perp/\mathbb{Z}F$ the class of this torus. The quasipolarisation $M \subset \mathrm{NS}(Y)$ is *compatible* with the fibration and γ [GT24, Definition 4.21] if $F \in M$ and $\bar\tau$ has a lift $\tau \in F^\perp \subset H^2(Y;\mathbb{Z})$ with $\tau \in M^\perp$. Given such a lift one forms

$$\Lambda_Y = (\mathbb{Z}F + \mathbb{Z}\tau)^\perp/(\mathbb{Z}F + \mathbb{Z}\tau), \qquad \Gamma = F_M^\perp/\mathbb{Z}F \subset \Lambda_Y, \qquad S_{Y,i} = \mathrm{im}\big(k_i^\perp \longrightarrow \Lambda_Y\big), \tag{3}$$

where $k_i^\perp \subset N_i$ and the maps come from the integral gluing of [GT24, Propositions 4.18 and 4.20], recalled in (23) below. The four clauses of the *mirror pair* definition [GT24, Definition 5.1] are verified one by one, each identified by its number, in §9.

2.4 Elliptic fibrations

Singular fibres are named by Kodaira type: I_1 (nodal, Euler number 1), II^* (Euler number 10, components not meeting the zero section spanning E_8), I_n^* (Euler number $n+6$, such components spanning D_{n+4}). A nodal fibre has monodromy the Picard–Lefschetz transvection T_v along its vanishing cycle $v \in H_1(F_p)$; in the basis (a, b), with $\langle a, b\rangle = -1$ and $v = pa + qb$,

$$T_{(p,q)} = \begin{pmatrix} 1-pq & p^2 \\ -q^2 & 1+pq \end{pmatrix}, \qquad \text{so that} \qquad T_a = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right). \tag{4}$$

3 The restriction period and very generality

The *restriction period* of the central fibre [AE23, Construction 4.3, Proposition 4.4] is the homomorphism $\varphi\colon \Lambda_X \to \mathrm{Pic}^0(C)$ sending the class of $(D_1, D_2) \in \xi^\perp$ to $D_1|_C - D_2|_C$. It is well defined: $D_1 \cdot c_1 = D_2 \cdot c_2$ on $\xi^\perp$, so the difference has degree zero, and $\varphi(\xi) = c_1|_C + c_2|_C = 0$ is exactly d-semistability. The inclusion $L \subset \ker\varphi$ holds because the classes of $\mathcal{L}$ extend to line bundles on all of $\mathcal{X}$, the period arc of $\mathcal{X}$ (the image of Δ^* under the marked period map)

lying in the H-locus (the locus of H-polarised periods) [AE23, §7E], and the restrictions of a line bundle on X_0 to V_1 and to V_2 agree on C. Since $L \simeq H$ is unimodular, $\Lambda_X = L \oplus R_X$ with R_X even, unimodular and negative definite of rank 16. Before using the period domain we lift this splitting to $H^2(X_t)$.

Lemma 3.1. *There is an orthogonal decomposition*

$$H^2(X_t;\mathbb{Z}) = H_{e_1} \oplus H_{e_2} \oplus L \oplus R_X, \qquad H_{e_j} = \mathbb{Z}e_j \oplus \mathbb{Z}e_j^\vee \simeq H, \quad I = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2, \tag{5}$$

compatible with (1)*, i.e. inducing the identification* $I^\perp/I = L \oplus R_X$*.*

Proof. The unimodular L splits off $H^2(X_t;\mathbb{Z})$; the complement $L^\perp$ is even unimodular and I is primitive in it, so restriction to I maps $L^\perp$ onto $I^\vee$. Choose a basis e_1, e_2 of I and $e_j^\vee \in L^\perp$ with $e_i \cdot e_j^\vee = \delta_{ij}$; replace $e_j^\vee$ by $e_j^\vee - \frac{1}{2}(e_j^\vee)^2 e_j$ (integral by evenness) and then $e_2^\vee$ by $e_2^\vee - (e_1^\vee \cdot e_2^\vee)e_1$. The new $e_j^\vee$ are isotropic and mutually orthogonal, so $H_{e_1} \oplus H_{e_2}$ is unimodular, contains I and splits off; its complement in $L^\perp$ maps isomorphically onto $(I^\perp/I)/L = R_X$. □

Lemma 3.2. *Under the very-generality hypothesis of Theorem 1.2,* $\ker\varphi = L$*.*

Proof. Because φ vanishes on L, it suffices to show that $\varphi(r) \neq 0$ for $0 \neq r \in R_X$. Fix (5), in which R_X is lifted to a sublattice of $H^2(X_t;\mathbb{Z})$ orthogonal to e_1, e_2. In a mixed-marked Type II chart of the period domain near the boundary point of the degeneration [AE23, Theorem 7.1], the coordinates are the modulus τ_C of $C \simeq \mathbb{C}/(\mathbb{Z} + \tau_C\mathbb{Z})$, an extension coordinate $z \in R_X \otimes \mathbb{C}$ and a logarithmic smoothing coordinate w (the boundary corresponds to $\operatorname{Im} w \to \infty$). In the decomposition (5), normalised by $(\Omega, e_1) = 1$, the period vector is $\Omega = e_1^\vee + \tau_C e_2^\vee + z + we_2 - (\tau_C w + z^2/2)e_1$, the last coefficient being forced by $\Omega^2 = 0$; and the restriction period $\varphi(r)$ of $r \in R_X$ is (r, z) mod $(\mathbb{Z} + \tau_C\mathbb{Z})$, the extension coordinate recording the period of the central fibre [AE23, Construction 4.3, Proposition 4.13]. If this vanishes for some $r \neq 0$, then $(r, z) = m + n\tau_C$ with $m, n \in \mathbb{Z}$, and $r' = r - me_1 - ne_2$ is integral, orthogonal to L, of square $r'^2 = r^2 < 0$ (as I is isotropic and orthogonal to r), with $(\Omega, r') = (r, z) - m - n\tau_C = 0$. So the hyperplane $\{(\Omega, r') = 0\}$ of periods orthogonal to r' passes through the boundary point and meets the interior of the period domain (its equation does not involve w, so take $\operatorname{Im} w$ large): the boundary point lies on the closure of the Noether–Lefschetz locus of r', whose image in the ABE compactification is the closed Noether–Lefschetz locus excluded by the hypothesis. □

Consequently R_X is isometric to E_8^2 or to D_{16}^+ [Se62, §8]; these are the two Type II boundary components ("cusps") of the H-polarised period space [ABE22, §7F, Theorem 7.58].

Finally we fix a basis of L. The lattice $L \simeq H$ contains exactly two roots $\pm s$; by Riemann–Roch one of them is effective on each smooth fibre, and the effective sign is constant in the smooth family (semicontinuity). Let s be the effective root and f the primitive isotropic class with $s \cdot f = 1$ lying in the closure of the positive cone of X_t (the component containing A_t); of the two candidates f and $-s - f$ exactly one does. Then

$$s^2 = -2, \qquad f^2 = 0, \qquad s \cdot f = 1, \qquad [A_t] = as + bf \quad \text{with } a > 0,\ b \geq 2a. \tag{6}$$

The inequalities hold because A_t is nef and big: $A_t \cdot f = a \geq 0$ as both lie in the closed positive cone, $A_t \cdot s = b - 2a \geq 0$ as s is effective, and $A_t^2 = 2a(b-a) > 0$ forces $a > 0$.

4 Reference models

The hypotheses give no explicit description of X_0. We therefore construct, for the given marked period, an explicit Kulikov model $\mathcal{W} \to \Delta$ with the same punctured family; §5 compares $\mathcal{X}$ with it. A *surface pair* is a d-semistable union $V_1 \cup_C V_2$ of smooth rational surfaces along a common anticanonical elliptic curve, and Λ_W denotes the lattice (1) of such a pair (it depends only on $\mathrm{NS}(V_i) \ni c_i$). An *interior* (-2)-curve on a component is one disjoint from C.

Proposition 4.1. *After shrinking the disc, there is a Kulikov model $\mathcal{W} \to \Delta$, isomorphic over Δ^* to the given family together with its H-classes and monodromy, without base change, whose central fibre $W_1 \cup W_2$ is as follows. If $R_X = E_8^2$, the W_i are rational elliptic surfaces without reducible fibres; the double curve is a fibre c_i of each, and there are sections $s_i \subset W_i$ ((-1)-curves) meeting C at the same point, with*

$$s = (s_1, s_2), \qquad f = (c_1, 0) = (0, c_2) \quad \text{in } \Lambda_W. \tag{7}$$

If $R_X = D_{16}^+$, then $W_1 = \mathbb{F}_2$ is the Hirzebruch surface with negative section s_1, ruling g_1, and $c_1 = 2s_1 + 4g_1$ (so $c_1^2 = 8$ and $C \cap s_1 = \emptyset$); W_2 is the blow-up of $\mathbb{F}_0$ at sixteen points of a smooth anticanonical curve of bidegree $(2,2)$, lying on distinct fibres of one ruling g_2; W_2 has no interior (-2)-curve, and

$$s = (s_1, 0), \qquad f = (g_1, g_2), \qquad c_2^2 = -8. \tag{8}$$

The birational identification $\mathcal{W} \dashrightarrow \mathcal{X}$ is an isomorphism in codimension one; in particular the divisor A itself has a strict transform on $\mathcal{W}$, an effective, flat Cartier divisor.

The construction realises the *given* φ, not just its cusp type (compare [AE23, Proposition 7.23]). It uses the lattices Λ_W of the two pairs, $L \oplus E_8^{(1)} \oplus E_8^{(2)}$ (the superscript indicating the component W_i carrying the summand) and $L \oplus D_{16}^+$, with s, f as in (7), (8), computed in §6 directly from $\mathrm{NS}(W_i)$ (independently of Lemma 5.1 below). As R_X is isometric to the rank-16 summand, we fix an isometry $\Lambda_W \to \Lambda_X$ sending s, f to s, f, and regard φ as a homomorphism on Λ_W, hence on $E_8^{(i)}$, vanishing on L.

Proof. Step 1: a surface pair with period φ, case E_8^2. Let $P_i = \mathbb{Z}l_i \oplus \bigoplus_{j=1}^{9} \mathbb{Z}e_{ij}$ be the Picard lattice of $\mathbb{P}^2$ blown up at nine points, with anticanonical class $c_i = 3l_i - \sum_j e_{ij}$ and $s_i = e_{i9}$. The plane $\langle s_i, c_i \rangle$ is unimodular (Gram matrix $\left(\begin{smallmatrix} -1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$), so $P_i = \langle s_i, c_i \rangle \oplus E_8^{(i)}$. Choose $P \in \mathrm{Pic}^1(C)$ and define $\rho_i \colon P_i \to \mathrm{Pic}(C)$ by $\rho_i(s_i) = P$, $\rho_i(c_i) = 0$, $\rho_1|_{E_8^{(1)}} = \varphi$, $\rho_2|_{E_8^{(2)}} = -\varphi$ (the sign comes from the definition of φ). Then $\deg \rho_i(D) = D \cdot c_i$. Embed C in a plane by the degree-three bundle $\rho_i(l_i)$ and blow up the nine points of C given by the degree-one

bundles $\rho_i(e_{ij})$. On the resulting surface W_i the curve C is anticanonical with restriction map ρ_i; as $\rho_i(c_i) = 0$, its normal bundle is trivial and $0 \to \mathcal{O} \to \mathcal{O}(C) \to \mathcal{O}_C \to 0$ shows that $|c_i|$ is a basepoint-free pencil with section s_i. A coincidence among the nine points, or a reducible fibre, would give a root in $c_i^\perp$ with zero period ($e_{ij} - e_{ik}$, respectively an interior (-2)-curve). Its class in $\Lambda_W = L \oplus R_X$ is a root outside L, since the classes of L supported on one component are the multiples of the isotropic f; as φ vanishes on L, its R_X-component would be a nonzero element of $\ker\varphi$, contradicting Lemma 3.2. Both sections meet C at P. This constructs (7) with restriction period φ.

Step 2: case D_{16}^+. Choose $G \in \mathrm{Pic}^2(C)$. The double cover $C \to \mathbb{P}^1$ defined by $|G|$ embeds C in $\mathbb{F}_2$ as an anticanonical curve disjoint from the negative section, with normal bundle $4G$. For the other component consider the abstract lattice $P_2 = \mathbb{Z}z \oplus \mathbb{Z}g_2 \oplus \bigoplus_{j=1}^{16} \mathbb{Z}E_j$ with $z^2 = g_2^2 = 0$, $z \cdot g_2 = 1$, $E_j^2 = -1$ and $c_2 = 2z + 2g_2 - \sum_j E_j$, the Picard lattice of $\mathbb{F}_0$ blown up at sixteen points, and its index-two sublattice $P_2^{\mathrm{ev}} = \{D : c_2 \cdot D \in 2\mathbb{Z}\}$. For $D \in P_2^{\mathrm{ev}}$ with $c_2 \cdot D = 2m$ prescribe $\rho_2(D) = mG - \varphi([(mg_1, D)])$, where $(mg_1, D) \in \xi^\perp$ because $mg_1 \cdot c_1 = 2m$. This is a homomorphism with $\rho_2(g_2) = G$ and $\rho_2(c_2) = -4G$, since $(g_1, g_2) = f$ and $(-4g_1, c_2) + \xi = 2s$ lie in $L = \ker\varphi$. Extend ρ_2 to P_2 by choosing a degree-one square root of $\rho_2(2E_1)$ (multiplication by two is surjective on $\mathrm{Pic}^0(C)$). Put $Z = \rho_2(z)$ and $p_j = \rho_2(E_j) \in C$. The degree-two bundles Z, G are distinct, since otherwise the root $z - g_2 \perp c_2$ would have zero period; their two maps to $\mathbb{P}^1$ embed C as a smooth $(2,2)$-curve in $\mathbb{F}_0$. Blow up the p_j. They are distinct and lie on distinct fibres of the ruling g_2, since otherwise $E_i - E_j$ or $g_2 - E_i - E_j$ would have zero period; these negative-square classes lie outside L, whose classes supported on this component are multiples of $2s + 4f$, of positive square. The resulting restriction map is ρ_2, and $\sum_j p_j = 2Z + 6G$ (the degree-sixteen part of $\rho_2(c_2) = -4G$) says that the normal bundle of C in W_2 is $-4G$: the pair is d-semistable. Its restriction period is φ by the defining formula on P_2^{ev}, and an interior (-2)-curve on W_2 would again contradict $\ker\varphi = L$. This is the generic component (8), not the special $\mathrm{Bl}_{16}\mathbb{F}_2$ model of [ABE22, §7F].

Step 3: a family over the same disc. A Tyurin degeneration has monodromy of multiplicity one in the sense of [AE23, Theorem 3.13], so its mixed-marked period arc is transverse to the boundary. Both constructed pairs are d-semistable, hence are central fibres of Kulikov models [Fr83], and by the Type II local period theorem [AE23, Theorem 7.1, Corollary 7.24] the local period map on the semiuniversal smoothing of such a pair is unramified. Pulling its local inverse back along the period arc of the *original* family gives a smooth Kulikov model $\mathcal{W}$ over the same disc, with no base change. The H-classes extend on the H-locus [AE23, §7E], and the markings can be matched at one smooth point, so the monodromy generators agree.

Step 4: identification over Δ^*. For an even integer $N \gg 0$ the class $A_N = 4s + Nf$ has central extensions $\big(4s_1 + \frac{N}{2}c_1,\ 4s_2 + \frac{N}{2}c_2\big)$, respectively $\big(2s_1 + (N-4)g_1,\ Ng_2 + c_2\big)$, on the two reference pairs; positivity on every curve follows from the pencils and the absence of interior roots, so these are ample. They extend on the H-locus and remain relatively ample near the reference central fibre.

On the original smooth fibres f is nef. Indeed, let $d = xs + yf + r$, $r \perp L$, be an effective

root with $x = d \cdot f < 0$. Nefness of A_t gives $0 \le A_t \cdot d = ay + (b-2a)x$, which with $x < 0$, $b \ge 2a$ forces $y \ge 0$; then $-2 = d^2 = -2x^2 + 2xy + r^2$ with $xy \le 0$, $r^2 \le 0$ forces $d = -s$, which is not effective. Since f lies in the closure of the positive cone and pairs nonnegatively with all effective roots, it is nef [Hu16, Ch. 8]. For $N/4 > b/a$, $A_N = (4/a)A_t + (N - 4b/a)f$ is nef on the original fibres, and ample because an A_N-orthogonal algebraic root there would be algebraic on the reference fibre with the same period, where A_N is ample. Strong Torelli [AE23, Theorem 2.3] now identifies the two punctured families with their H-classes, and the elementary-modification theorem for Kulikov models [SB83, Corollary 3.1] makes this identification an isomorphism in codimension one.

Finally, strict transform under a codimension-one isomorphism preserves horizontal prime divisors, and smoothness of $\mathcal{W}$ makes their sum Cartier and flat. All choices used the marked period of the given arc, never a coarse boundary point (§1.2). □

5 Classification of the central fibre

We now determine the central fibre of $\mathcal{X}$ itself. The idea is to transport A to $\mathcal{W}$, make it nef by at most one elementary modification, and compare the contractions of $\mathcal{W}$ and $\mathcal{X}$ defined by the common relative section ring $\bigoplus_{m \ge 0} \pi_* \mathcal{O}(mA)$. Componentwise positivity of A is what excludes the endpoints of the parameter intervals below.

Lemma 5.1. *Up to interchanging the components, the tuple $(\Lambda_X, L, S_{X,1}, S_{X,2})$ of $\mathcal{X}$ is that of one of the following surface pairs: (E_0) the reference pair (7); (E_1) the pair obtained from (7) by contracting s_1 in W_1 and blowing up the point $s_1 \cap C = s_2 \cap C$ on W_2, so that $c_1^2 = 1$, $c_2^2 = -1$; (D) the reference pair (8), with $c_1^2 = 8$ and $\mathrm{NS}(W_1)$ even.*

Proof. Step 1: the restrictions of A. Transfer the divisor A to $\mathcal{W}$ by Proposition 4.1; its restrictions D_1, D_2 to W_1, W_2 are effective. Two extensions of the line bundle $\mathcal{O}(A)$ from the punctured family differ by a multiple of a central component, hence (D_1, D_2) is determined by (6) up to adding an *integral* multiple of ξ.

In case (7) this gives, for an integer parameter,

$$D_1 = as_1 + xc_1, \qquad D_2 = as_2 + yc_2, \qquad x + y = b. \tag{9}$$

Since $s_i + c_i$ is nef on W_i and $(s_i + c_i) \cdot D_i$ equals x, respectively y, effectivity gives $x, y \ge 0$.

In case (8), for an integer u,

$$D_1 = (a - 2u)s_1 + (b - 4u)g_1, \qquad D_2 = bg_2 + uc_2. \tag{10}$$

Intersecting with the nef rulings gives $D_1 \cdot g_1 = a - 2u \ge 0$ and $D_2 \cdot g_2 = 2u \ge 0$, that is, $0 \le u \le a/2$.

Step 2: a nef model, case E_8^2. If $x, y \ge a$, both divisors in (9) are nef, and the only curves with zero intersection are the sections s_1 (if $x = a$) and s_2 (if $y = a$). If $0 \le x < a$, then $y > a$, and we perform the elementary modification along s_1: this (-1)-curve of W_1 meets C once, so it is a $(-1,-1)$-curve of the threefold; flopping it contracts s_1 in W_1 and blows

up its point of attachment on W_2. Write $q\colon W_2' \to W_2$ for this blow-up, E for its exceptional curve, W_1' for the contraction, and primes for strict transforms. The strict transform of A restricts to

$$D_1' = xc_1', \qquad D_2' = q^*(as_2 + yc_2) + (x-a)E, \tag{11}$$

the coefficient of E being forced by $D_1' \cdot c_1' = D_2' \cdot c_2' = x$. The pullback of c_1' to W_1 is $s_1 + c_1$, whose only curve of zero intersection is s_1; thus c_1' is ample on W_1' and D_1' is nef, with no curve of zero intersection when $x > 0$. On W_2', $D_2' \cdot E = a - x$, $D_2' \cdot s_2' = b - 2a$ and $D_2' \cdot c_2' = x$. For every other irreducible curve $\Theta \subset W_2$, of multiplicity m at the centre, $c_2 \cdot \Theta \geq m$ and $s_2 \cdot \Theta \geq 0$, hence $D_2' \cdot \Theta' = a(s_2 \cdot \Theta) + y(c_2 \cdot \Theta) - (a-x)m \geq (y - a + x)m \geq 0$; if $c_2 \cdot \Theta = 0$ then Θ is an irreducible fibre and $s_2 \cdot \Theta = 1 > 0$. So D_2' is nef, and for $0 < x < a$ the only possible curve of zero intersection is s_2', an interior (-2)-curve, occurring when $b = 2a$. The case $y < a$ is symmetric under interchanging the components.

Step 3: a nef model, case D_{16}^+. For $0 \leq u \leq a/2$ both divisors (10) are already nef: on $\mathbb{F}_2$, a class $\alpha s_1 + \beta g_1$ with $\alpha \geq 0$ is nef iff $\beta \geq 2\alpha$, and $b - 4u \geq 2(a - 2u)$ is $b \geq 2a$; on W_2, every irreducible curve other than C has nonnegative intersection with both g_2 and c_2, while $D_2 \cdot C = 2b - 8u \geq 4a - 8u \geq 0$. The squares are $D_1^2 = 2(a-2u)(b-a-2u)$ and $D_2^2 = 4u(b-2u)$. For $0 < u < a/2$ the only possible curve of zero intersection is s_1 (an interior (-2)-curve), when $b = 2a$: indeed $D_1 \cdot s_1 = b - 2a$, $D_1 \cdot g_1 = a - 2u > 0$, $D_2 \cdot C > 0$, and each component of a singular fibre of the ruling g_2 has positive c_2-degree, since the blown-up points lie on distinct fibres.

Step 4: the common contraction excludes the endpoints. Denote the nef model of Steps 2–3 by $(\mathcal{W}', D)$; it is isomorphic to $\mathcal{X}$ in codimension one, so by Hartogs extension the relative section rings of $(\mathcal{W}', D)$ and $(\mathcal{X}, A)$ coincide. A relatively nef extension of a line bundle that is nef and big on the general fibre is relatively semiample [ABE22, §2A], so there are proper bimeromorphic morphisms $\mathcal{W}' \xrightarrow{\mu} \mathcal{Z} \xleftarrow{\nu} \mathcal{X}$ to the same normal relative Proj. A central component is contracted to dimension at most one precisely when the restriction of the nef line bundle to it is not big, and the components of $\mathcal{W}_0'$ and X_0 correspond under the codimension-one identification. At $x = 0$, $y = 0$, $u = 0$ or $u = a/2$ one of the D_i has square zero, so the corresponding V_i would be contracted, contradicting $(A|_{V_i})^2 > 0$.

Step 5: resolution counting. It remains to see that the remaining nef models determine X_0; we do this directly, without assuming connectivity of nef models by divisor-trivial flops. First, ν does not contract the double curve of X_0: the squares of C on V_1, V_2 sum to zero, so one is nonnegative, and the Hodge index theorem for the nef and big class $A|_{V_i}$ gives $A|_{V_i} \cdot C > 0$ there (a nef and big class orthogonal to C would force $C^2 < 0$); the two degrees $A|_{V_i} \cdot C$ agree, both being $\deg A|_C$. Let Z_i be the normalisations of the central components of $\mathcal{Z}$. By the lists of curves of zero intersection in Steps 2–3, the minimal resolutions $U_i \to Z_i$ are obtained from W_i' by contracting the (-1)-sections of zero D-degree, while interior (-2)-curves of zero degree are retained. The double curves $C_i^U \subset U_i$, identified with C, are smooth with $K_{U_i} + C_i^U = 0$. Each birational morphism $V_i \to Z_i$ factors through the minimal resolution U_i, and $V_i \to U_i$ is a composition of n_i point blow-ups. The identity $K_{V_i} + C = 0$ forces every centre to lie on the successive strict transform of C_i^U: a blow-up off that curve introduces a positive discrepancy which later blow-ups cannot cancel. Each such blow-up

lowers C^2 by one, so d-semistability of X_0 gives $n_1+n_2 = (C_1^U)^2+(C_2^U)^2$. Moreover, no centre lies over a smooth point of $\mathcal{Z}$: a proper bimeromorphic morphism from a smooth space with trivial canonical bundle to a normal space is an isomorphism over the smooth locus of the target. (Over a smooth ball $U \subset \mathcal{Z}$, the Jacobian of ν with respect to nowhere vanishing volume forms is a holomorphic function on $\nu^{-1}(U)$ vanishing exactly on the exceptional locus of ν; it descends through $\nu_*\mathcal{O}_\mathcal{X} = \mathcal{O}_\mathcal{Z}$ to a holomorphic function on U, whose zero set is the image of that exceptional locus, of codimension at least two in U, hence empty.) Note that only proper analytic contractions and projective surface resolutions have been used; projectivity of $\mathcal{X}$ is not assumed.

Now count. If no (-1)-section was contracted, the right-hand side of this identity is $(C^2)_{W_1'} + (C^2)_{W_2'} = 0$, so $V_i = U_i = W_i'$: this gives E_0 (for $x, y > a$), E_1 (for $0 < x < a$ or $0 < y < a$) and D. If exactly one section was contracted, say s_1 with $x = a < y$, the right-hand side is one; away from the image p_0 of s_1, μ is an isomorphism along the double curve, so $\mathcal{Z}$ is smooth there, and the single blow-up is at the point over p_0, on U_1 (giving back W_1, i.e. E_0) or on U_2 (giving E_1). Two sections are contracted only when $x = y = a$, so $b = 2a$; they meet C at the same point, the count is two, both centres lie over p_0, and a second blow-up on the same component must be at the unique point of the strict transform of the double curve over p_0. The distributions $(1,1), (2,0), (0,2)$ of the two blow-ups give E_0 and the two orders of E_1.

Step 6: the embedded lattices. Thus the central pair $(V_1 \cup V_2, C)$ is isomorphic to one of the listed pairs. The construction (1)–(2) is intrinsic to the central pair, the isomorphism preserves the restriction period, and $L = \ker\varphi$ by Lemma 3.2; hence it identifies the *given* embedded L and both $S_{X,i}$ with those of the listed pair, without any choice of a full cohomology marking. □

6 The embedded lattices on the degeneration side

We compute the tuple (Λ, L, S_1, S_2) of (1), (2) for each of the three surface pairs of Lemma 5.1 directly from the pair; by that lemma the result is the tuple of $\mathcal{X}$. In the splitting $L \oplus R_X$, with the basis s, f of L of (6), one finds the following; labels $1, 2$ may be interchanged here and below.

$$\mathrm{E}_0: \quad R_X = E_8^{(1)} \oplus E_8^{(2)}, \qquad S_{X,i} = \mathbb{Z}f \oplus E_8^{(i)}; \tag{12}$$

$$\mathrm{E}_1: \quad R_X = E_8^{(1)} \oplus E_8^{(2)}, \qquad S_{X,1} = E_8^{(1)}, \qquad S_{X,2} = L \oplus E_8^{(2)}. \tag{13}$$

For the third case write $h = s + 2f$ (so $h^2 = 2$) and $\epsilon\colon D_{16}^+ \to D_{16}^+/D_{16} \simeq \mathbb{Z}/2$; then

$$\mathrm{D}: \quad R_X = D_{16}^+, \qquad S_{X,1} = \mathbb{Z}s, \qquad S_{X,2} = \{\, mh+r : m \in \mathbb{Z},\ r \in D_{16}^+,\ \epsilon(r) \equiv m \pmod 2 \,\}. \tag{14}$$

Case E_0. Here $\mathrm{NS}(W_i) = \mathbb{Z}s_i \oplus \mathbb{Z}c_i \oplus E_8^{(i)}$ with $\left(\begin{smallmatrix} s_i^2 & s_ic_i \\ s_ic_i & c_i^2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} -1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, and $c_i^\perp = \mathbb{Z}c_i \oplus E_8^{(i)}$. Passing to $\xi^\perp/\mathbb{Z}\xi$, where $(c_1, 0) = (0, c_2) = f$ and $(s_1, s_2) = s$, gives (12).

Case E_1. Contracting s_1 leaves $c_1^\perp = E_8^{(1)}$ unchanged (now $c_1^2 = 1$), so $S_{X,1} = E_8^{(1)}$, a unimodular summand of Λ_X. On the other component $c_2^2 = -1$, so $\mathbb{Z}c_2$ splits off $\mathrm{NS}(V_2)$ and $c_2^\perp$ is unimodular of rank 10; its image $S_{X,2}$ is a unimodular sublattice of the rank-10 lattice $S_{X,1}^\perp = L \oplus E_8^{(2)}$, hence equal to it. This proves (13).

Case D. We first locate the root sublattice $D_{16} \subset R_X$. In the basis z, g_2, E_j of $\mathrm{NS}(W_2) = P_2$ from the proof of Proposition 4.1, the lattice

$$R_0 = c_2^\perp \cap g_2^\perp = \Big\{ -\tfrac{1}{2}\big(\textstyle\sum_j t_j\big) g_2 + \sum_j t_j E_j : t_j \in \mathbb{Z},\ \sum_j t_j \equiv 0 \pmod 2 \Big\} \tag{15}$$

has form $-\sum t_j^2$ on the even-sum vectors (t_j), so it is D_{16}. An element of $R_X = L^\perp$ has a representative

$$\big(k(s_1 + 2g_1), D\big), \qquad D \cdot g_2 = -k, \quad D \cdot c_2 = 4k, \tag{16}$$

since orthogonality to $s = (s_1, 0)$ forces the first entry to be a multiple of $s_1 + 2g_1$, orthogonality to f gives $D \cdot g_2 = -k$, and $(s_1 + 2g_1) \cdot c_1 = 4$. Changing the representative by $\xi = (2s_1 + 4g_1, -c_2)$ changes k by 2. The classes with even k form (15), and odd k occurs ($k = 1$, $D = -z + 3g_2$), so R_X is an even unimodular index-two overlattice of D_{16}: $R_X = D_{16}^+$, with ϵ the parity of k. Next, $c_1^\perp = \mathbb{Z}s_1$ in $\mathrm{NS}(\mathbb{F}_2)$, so $S_{X,1} = \mathbb{Z}s$. If $(0, D)$ with $D \cdot c_2 = 0$ represents an element of $S_{X,2}$, put $m = D \cdot g_2$; subtracting $mh = (m(s_1 + 2g_1), 2mg_2)$ leaves the element $(-m(s_1 + 2g_1), D - 2mg_2)$ of R_X, whose coefficient in (16) is $k = -m$, so $\epsilon(r) \equiv m$. Conversely, given $mh + r$ with $\epsilon(r) \equiv m$, choose the representative of r with $k = -m$; then $mh + r = (0, D + 2mg_2)$ lies in $S_{X,2}$. This proves (14).

7 Fibrations and splitting loops on Y

Since Y is very general, $\mathrm{NS}(Y) = M$. We use two elliptic fibrations with section on Y: the *standard* fibration, with singular fibres $2\,\mathrm{II}^* + 4\,\mathrm{I}_1$, and the *alternate* fibration, with singular fibres $\mathrm{I}_{12}^* + 6\,\mathrm{I}_1$ [DPT25, §2.2], [CD07].

The standard fibration and the loops γ_0, γ_1. On the base with coordinate $t \neq 0, \infty$, a Weierstrass equation of the standard fibration can be normalised to the Inose form [CD07]

$$z^2 = u^3 + \alpha u + \beta + t + t^{-1} : \tag{17}$$

the II^* conditions at $t = 0, \infty$ force the Weierstrass coefficients to be αt^4 and $t^5(\lambda_0 + \lambda_1 t + \lambda_2 t^2)$, and rescaling t, u, z gives (17), with generic parameters α, β. We use two splitting loops, distributing the singular fibres over the two pieces $Y_1 \mid Y_2$ as

$$\gamma_0 \colon\ \mathrm{II}^* + 2\,\mathrm{I}_1 \ \big|\ \mathrm{II}^* + 2\,\mathrm{I}_1, \qquad\qquad \gamma_1 \colon\ \mathrm{II}^* + \mathrm{I}_1 \ \big|\ \mathrm{II}^* + 3\,\mathrm{I}_1. \tag{18}$$

Since arbitrary loops with these fibre counts need not have the monodromy required for allowability, we construct them explicitly. The standard fibration is the quadratic base change $v = \beta + t + t^{-1}$ of the rational elliptic surface $z^2 = u^3 + \alpha u + v$, whose singular fibres

are $\mathrm{II}^* + 2\,\mathrm{I}_1$. Choose the parameters so that some disc containing the two branch values $v = \beta \pm 2$ contains no critical value of this rational surface. Over the complement of that disc the cover is trivial, a disjoint union of two copies of the complement; the boundary of either copy is γ_0, with identity monodromy on each side. Moving one nodal critical value across this boundary gives γ_1: choosing the symplectic basis so that its vanishing cycle is a, the anticlockwise monodromy on the side with Euler number 11 is T_a^{-1} and on the other side it is T_a. These constructions transport to the given generic parameters: the space of base changes with distinct branch and critical values is connected, and the fibrations and loops are topologically locally trivial over it.

The alternate fibration and the loop γ_D. Taking u as base coordinate in (17) and $\eta = tz$ gives the alternate fibration

$$\eta^2 = t\big(t^2 + P(u)t + 1\big), \qquad P(u) = u^3 + \alpha u + \beta, \tag{19}$$

with a section, a section of order two ($t = \eta = 0$), an I_{12}^* fibre at $u = \infty$ (the Weierstrass invariant c_4 and the discriminant vanish there to orders 2 and 18) and six nodal fibres over $P(u) = \pm 2$ (the discriminant away from infinity is a nonzero constant times $P(u)^2 - 4$). We construct a loop γ_D with the splitting $4\,\mathrm{I}_1 \mid \mathrm{I}_{12}^* + 2\,\mathrm{I}_1$. View (19) as the pull-back under the cubic $v = P(u)$ of the rational elliptic surface $\eta^2 = t(t^2 + vt + 1)$, with singular fibres $\mathrm{I}_4^* + 2\,\mathrm{I}_1$. On this rational surface the two nodal vanishing cycles may be taken to be

$$B = b, \qquad D = b + 2a \qquad (\langle a, b\rangle = -1)\,: \tag{20}$$

the product of the two positive nodal Picard–Lefschetz transformations is the inverse of the I_4^* monodromy, of trace -2; as $T_B T_D$ has trace $2 - \langle B, D\rangle^2$, the vanishing cycles have intersection number ± 2, and a choice of signs and of the symplectic basis gives (20).

Choose the cubic P so that the inverse image of a disc containing ± 2 consists of three disjoint discs. A path in the v-plane from that disc to a finite branch value of P lifts to an arc joining two of the three discs; the two finite branch values of the generic cubic P connect in this way two different pairs of discs (the cover is connected), hence all three. The image of each connecting arc goes out and back through regular values of the rational elliptic surface, so parallel transport along it is trivial on $H_1(F_p)$. Thus, in a common marking, the vanishing-cycle pairs in the three discs are $(B, D), (B, D), (B, D)$. A regular neighbourhood of two of the discs and their connecting arc is a disc; its boundary is γ_D. As before, connectedness of the complement of the discriminant in the parameter space of cubics transports this loop to all generic members of (19). In the convention (4),

$$T_B T_D = \left(\begin{smallmatrix} -1 & 4 \\ 0 & -1 \end{smallmatrix}\right), \qquad (T_B T_D)^2 = \left(\begin{smallmatrix} 1 & -8 \\ 0 & 1 \end{smallmatrix}\right) = T_a^{-8}, \tag{21}$$

which is the monodromy around γ_D on the four-node side, as required for $e(Y_1) = 4$; the complementary monodromy is its inverse T_a^8, as required for $e(Y_2) = 20$.

Allowability. The monodromies computed above are $\left(\begin{smallmatrix}1 & e(Y_i)-12\\ 0 & 1\end{smallmatrix}\right)$ in all cases: the Euler numbers are $12, 12$ for γ_0, $11, 13$ for γ_1 and $4, 20$ for γ_D. It remains to check primitivity of $r_i(a)$. An integral relative class v of charge b, i.e. with $\phi_i(v) = b$, satisfies $\chi_i(v, r_i(a)) = \langle b, a\rangle = 1$, so $r_i(a)$ is primitive whenever such a v exists. For the splittings (18) the charge map is surjective already on a neighbourhood of the II^* fibre, since its nodal resolution has charges generating $H_1(F_p)$ [GT24, Lemma 4.7, Proposition 4.8, Table 4.1], so this applies for every chosen primitive a. For γ_D each side contains a nodal pair (20), so it applies directly. Consequently all three loops are allowable in the sense of [GT24, Definition 4.11].

8 Compatibility and the lattices on the fibration side

Compatibility. For any of the three loops, the torus class $\bar\tau \in F^\perp/\mathbb{Z}F$ of §2.3 is primitive and isotropic by the integral gluing description [GT24, Propositions 4.18 and 4.20]. Choose a lift $\tau_0 \in F^\perp$, let Q be the class of the zero section, and set $\tau = \tau_0 - (\tau_0 \cdot Q)F$. The torus lies over the loop, hence in smooth fibres, so τ_0 is orthogonal to F and to every singular-fibre component, and the correction makes $\tau \cdot Q = 0$ as well. These classes span $\mathrm{NS}(Y) \otimes \mathbb{Q}$ (Q, F and the fibre components not meeting Q span a sublattice of rank $18 = \operatorname{rk} M$ in both fibrations), hence $F \in M$ and $\tau \in M^\perp$. This is compatibility, including the integral lift required by [GT24, Definition 4.21].

The frame. With Λ_Y and Γ as in (3), since M is unimodular and $Q \cdot F = 1$,

$$\Lambda_Y = H_0 \oplus \Gamma, \qquad H_0 = \tau_{M^\perp}^\perp/\mathbb{Z}\tau \simeq H, \qquad \Gamma \simeq E_8^{(1)} \oplus E_8^{(2)} \text{ (standard)}, \quad \Gamma \simeq D_{16}^+ \text{ (alternate)}. \tag{22}$$

Indeed $H^2(Y;\mathbb{Z}) = M \oplus M^\perp$ with $F \in M$, $\tau \in M^\perp$, so $\Lambda_Y = \Gamma \oplus (\tau_{M^\perp}^\perp/\mathbb{Z}\tau)$ orthogonally (whence $H_0 = \Gamma^\perp$ in Λ_Y), and the second summand is H because $M^\perp \simeq H^{\oplus 2}$ and τ is primitive isotropic (primitivity of $\bar\tau$ implies that of τ). The unimodular plane $\langle Q, F\rangle \simeq H$ splits off M, so Γ is even unimodular negative definite of rank 16 containing the fibre-root lattice $E_8 \oplus E_8$, respectively D_{16}; in the alternate case D_{16} has index two in Γ, by its determinant 4 (or by the two-torsion section).

Pseudolattice facts. The gluing results of [GT24, Propositions 4.18 and 4.20] give the integral identification

$$\Lambda_Y = (k_1, -k_2)^\perp/\mathbb{Z}(k_1, -k_2) \;\subset\; (N_1 \oplus N_2)/\mathbb{Z}(k_1, -k_2), \tag{23}$$

with $S_{Y,i}$ the image of $k_i^\perp \subset N_i$. The two images are orthogonal, the maps $k_i^\perp \to \Lambda_Y$ are injective (no $(0, w)$ with $w \neq 0$ is a multiple of $(k_1, -k_2)$), and the classes of fibre components of Y_i go to the corresponding fibre roots in Γ. For an allowable piece of Euler number $n \in \{11, 12, 13, 20\}$ the lattice N_i is the odd unimodular lattice $\mathrm{I}_{1,n-3}$, and k_i is characteristic, primitive and of square $12 - n$ [GT24, Theorems 2.18 and 4.10, Lemma 2.20]; for the pieces with a II^* fibre, primitivity also follows from the surjectivity of ϕ_i noted in §7.

The sign convention for k_i does not affect $k_i^\perp$. As at least one k_i is primitive in each case, $(k_1, -k_2)$ spans a saturated subgroup and (23) is the integral construction, not a rational one. The piece of Euler number 4 is computed by hand below.

The splitting γ_0. Here $e(Y_i) = 12$, so k_i is primitive, isotropic and characteristic in $N_i = \mathrm{I}_{1,9}$. The fibre roots of the II^* fibre on each side give a unimodular $E_8^{(i)} \subset k_i^\perp$, which splits off: $N_i = E_8^{(i)} \oplus U_i$ with $U_i \ni k_i$ odd unimodular of signature $(1,1)$. Choose $\varepsilon_i \in U_i$ with $\varepsilon_i \cdot k_i = 1$ (unimodularity and primitivity); since k_i is characteristic, ε_i^2 is odd, and adding a multiple of k_i we may assume $\varepsilon_i^2 = -1$. Thus $N_i = (\mathbb{Z}\varepsilon_i \oplus \mathbb{Z}k_i) \oplus E_8^{(i)}$ and $k_i^\perp = \mathbb{Z}k_i \oplus E_8^{(i)}$. In (23) the images of $(k_1, 0)$ and $(0, k_2)$ coincide in a primitive isotropic vector κ, and together with the image of $(\varepsilon_1, \varepsilon_2)$ (square -2, pairing 1 with κ) it spans $H_0 = \Gamma^\perp$. Hence $S_{Y,i} = \mathbb{Z}\kappa \oplus E_8^{(i)}$. Mapping $\kappa \mapsto f$, $H_0 \to L$ isometrically and the root summands to their namesakes gives an isometry $\psi\colon \Lambda_Y \to \Lambda_X$ carrying (22) and the $S_{Y,i}$ to (12), with $\Gamma \mapsto R_X$.

The splitting γ_1. Here $k_1^2 = 1$ and $k_2^2 = -1$, so $\mathbb{Z}k_i$ splits off N_i. The complement $k_1^\perp$ is unimodular of rank 8 and contains the fibre roots $E_8^{(1)}$, hence equals them. Writing $N_1 = \mathbb{Z}k_1 \oplus E_8^{(1)}$ and $N_2 = \mathbb{Z}k_2 \oplus k_2^\perp$, the vectors of $(k_1, -k_2)^\perp$ are $\alpha(k_1, -k_2) + (e, w)$ with $e \in E_8^{(1)}$, $w \in k_2^\perp$, so (23) splits integrally as $\Lambda_Y = E_8^{(1)} \oplus k_2^\perp$, with $S_{Y,1} = E_8^{(1)}$ and $S_{Y,2} = k_2^\perp = H_0 \oplus E_8^{(2)}$. The last equality holds because $k_2^\perp$ is unimodular of signature $(1,9)$, even (as k_2 is characteristic), and contains the unimodular fibre roots $E_8^{(2)}$. This is (13), and any isometry $H_0 \to L$ together with the identity on the root summands gives ψ, again with $\Gamma \mapsto R_X$.

The splitting γ_D. Two integral points must be checked: the small side has "degree $8'$" rather than 8 (its pseudolattice is that of $\mathbb{P}^1 \times \mathbb{P}^1$, with $N_1 \simeq H$ even and $k_1^\perp \simeq \langle -2 \rangle$, not that of $\mathbb{F}_1$, with $N_1 = \mathrm{I}_{1,1}$ and $k_1^\perp \simeq \langle -8 \rangle$), and the large side carries the parity condition of (14).

Small side. Use distinguished thimbles $\theta_1, \dots, \theta_4$ with charges B, D, B, D as in (20). In the basis (a, b), with $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, the charge matrix, the Seifert matrix and the right adjoint are

$$V = \begin{pmatrix} 0 & 2 & 0 & 2 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \qquad \chi_1 = \begin{pmatrix} 1 & 2 & 0 & 2 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad r_1 = \chi_1^{-1} V^t J,$$

the entries of $V^t J$ being $\langle \phi_1(\theta_i), a \rangle$ and $\langle \phi_1(\theta_i), b \rangle$. Direct multiplication gives $r_1(a) = (1, -1, -1, 1)$ and $r_1(b) = (-8, 6, 4, -2)$. Since $\chi_1(\theta_i, r_1(a)) = 1$ for all i, the subgroup $r_1(a)^\perp$ consists of the vectors whose coordinates sum to zero. Modulo $r_1(a)$ it has basis $u = \theta_1 - \theta_4$, $v = \theta_2 - \theta_4$, with $\big(q_1(u,u), q_1(u,v), q_1(v,v)\big) = (0, -1, -2)$ for $q_1 = -\chi_1$, and $k_1 = -4u + 2v$. In particular $N_1 \simeq H$ is even, $k_1^2 = 8$, and $k_1^\perp = \mathbb{Z}v$ with $v^2 = -2$.

Write δ for the image of v in Λ_Y. Every D_{16} fibre-root class is supported on the other side of the loop, so is orthogonal to δ; the saturation of D_{16} is Γ, so $\delta \in \Gamma^\perp = H_0$, and it is a primitive root there. Choose a basis δ, f_0 of H_0 with $f_0^2 = 0$, $\delta \cdot f_0 = 1$, and put $h_0 = \delta + 2f_0$. Then

$$S_{Y,1} = \mathbb{Z}\delta, \qquad \delta^\perp_{\Lambda_Y} = \mathbb{Z}h_0 \oplus D_{16}^+. \tag{24}$$

Large side. Here $N_2 = \mathrm{I}_{1,17}$ and k_2 is primitive of square -8, so $k_2^\perp \simeq S_{Y,2}$ has determinant 8 and cyclic discriminant group $\mathbb{Z}/8$. Since $S_{Y,2} \perp S_{Y,1}$ and $\delta^\perp$ has determinant 2, $S_{Y,2}$ has index two in the right-hand side of (24). We claim that

$$S_{Y,2} \cap \Gamma = D_{16}. \tag{25}$$

Let G_* be the relative homology of a neighbourhood of the I_{12}^* fibre and order the two remaining nodal thimbles last, so $H_2(Y_2, F_p) = G_* \oplus \mathbb{Z}^2$. The kernel of the charge map on $G_* = H_2(U, F_p)$, U the neighbourhood, is the image of $H_2(U)$, that is $H_2(U)/\mathbb{Z}[F]$, which is spanned by the fibre components modulo the total fibre: it is exactly the D_{16} fibre-root lattice and is primitive in G_*. The two nodal thimbles have charges B, D with $\langle B, D\rangle = 2$, as on the small side, so the last diagonal block of the upper-triangular Seifert matrix is $\left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$ and the corresponding right-adjoint equation has right-hand side $(1,1)^t$; hence the last two coordinates of $r_2(a)$ are $(-1, 1)$. Hence D_{16} remains primitive in $N_2 = r_2(a)^\perp/\mathbb{Z}r_2(a)$: if $nx = d + m\, r_2(a)$ with $d \in D_{16}$, the last two coordinates show $n \mid m$, and primitivity in G_* gives $x - (m/n)r_2(a) \in D_{16}$. So D_{16} is primitive in $S_{Y,2}$, and its saturation $S_{Y,2} \cap \Gamma$ equals D_{16}, proving (25).

Thus $S_{Y,2}$ is the kernel of a nonzero character $\alpha\colon \mathbb{Z}h_0 \oplus D_{16}^+ \to \mathbb{Z}/2$ whose restriction to D_{16}^+ is ϵ, by (25). If $\alpha(h_0) = 0$, the kernel would be $\mathbb{Z}h_0 \oplus D_{16}$, with discriminant group $(\mathbb{Z}/2)^3$, not $\mathbb{Z}/8$. Hence $\alpha(h_0) = 1$ and

$$S_{Y,2} = \{\, mh_0 + r :\ \epsilon(r) \equiv m \pmod 2 \,\}. \tag{26}$$

Equations (22), (24) and (26) agree exactly with (14). Map δ, f_0 to s, f and identify the two copies of D_{16}^+ by any isometry (every isometry of D_{16}^+ preserves its root sublattice D_{16}, hence ϵ). This supplies the integral isometry in the third case.

In all three cases we have constructed an isometry $\psi\colon \Lambda_Y \to \Lambda_X$ with $\psi(\Gamma) = L^\perp_{\Lambda_X} = R_X$ and $\psi(S_{Y,i}) = S_{X,i}$ $(i = 1, 2)$, for the pairing of loops with central fibres listed in §1.3.

9 Lifting and proof of Theorem 1.2

It remains to verify the clauses of [GT24, Definition 5.1]. On the degeneration side we have (5). On the fibration side, τ is primitive and isotropic in $M^\perp \simeq H^{\oplus 2}$; in an even unimodular lattice a primitive isotropic vector has an isotropic dual pairing to 1, spanning with it a hyperbolic plane H_τ. Likewise $F \in M$ with dual $Q + F$ spans H_F. Hence $H^2(Y;\mathbb{Z}) = H_\tau \oplus H_F \oplus H_0 \oplus \Gamma$ with $M^\perp = H_\tau \oplus H_0$ and $M = H_F \oplus \Gamma$. The inclusions $L \subset \Lambda_X$ and $\Gamma \subset \Lambda_Y$ are split, in particular primitive: this is clause (1). Clause (2) is the componentwise

equality $\psi(S_{Y,i}) = S_{X,i}$ established in §8. For clause (3), extend ψ from $H_0 \oplus \Gamma$ by isometries $H_\tau \to H_{e_1}$ and $H_F \to H_{e_2}$ taking τ to e_1 and F to e_2; this gives

$$\widehat{\psi}\colon H^2(Y;\mathbb{Z}) \longrightarrow H^2(X_t;\mathbb{Z}), \qquad \widehat{\psi}(M^\perp) = H_{e_1} \oplus L, \tag{27}$$

since $\psi(H_0) = \psi(\Gamma^\perp) = R_X^\perp = L$. Finally, τ is 1-admissible in the sense of [GT24, Definition 3.30]: it is primitive and isotropic in $M^\perp$, and its isotropic dual $g \in M^\perp$ satisfies $(\tau, g) = 1$, so τ and g both have divisibility one in $M^\perp$. Thus (27) is clause (4a) with $m = 1$.

Proof of Theorem 1.2. By Lemma 5.1 the degeneration is of type E_0, E_1 or D, with embedded lattices (12), (13) or (14). Choose on the given very general Y the standard fibration with γ_0, the standard fibration with γ_1, or the alternate fibration with γ_D, respectively. These are allowable (§7), the M-quasipolarisation is compatible (§8), and ψ, (27) verify [GT24, Definition 5.1]. Hence $\mathcal{X} \to \Delta$ and $(Y \to \mathbb{P}^1, \gamma)$ form a mirror pair. □

References

[ABE22] V. Alexeev, A. Brunyate and P. Engel, *Compactifications of moduli of elliptic K3 surfaces: stable pair and toroidal*, Geom. Topol. **26** (2022), 3525–3588.

[AE23] V. Alexeev and P. Engel, *Compact moduli of K3 surfaces*, Ann. of Math. **198** (2023), 727–789.

[CD07] A. Clingher and C. F. Doran, *Modular invariants for lattice polarized K3 surfaces*, Michigan Math. J. **55** (2007), 355–393.

[Do96] I. V. Dolgachev, *Mirror symmetry for lattice polarized K3 surfaces*, J. Math. Sci. **81** (1996), 2599–2630.

[DHT17] C. F. Doran, A. Harder and A. Thompson, *Mirror symmetry, Tyurin degenerations and fibrations on Calabi–Yau manifolds*, Proc. Sympos. Pure Math. **96** (2017), 93–131.

[DPT25] C. F. Doran, J. Prebble and A. Thompson, *Normal forms and Tyurin degenerations of K3 surfaces polarised by a rank 18 lattice*, Math. Nachr. **298** (2025), 806–828.

[Fr83] R. Friedman, *Global smoothings of varieties with normal crossings*, Ann. of Math. **118** (1983), 75–114.

[GT24] L. Giovenzana and A. Thompson, *Degenerations and fibrations of K3 surfaces: lattice polarisations and mirror symmetry*, arXiv:2405.12009v3 (2024).

[HT20] A. Harder and A. Thompson, *Pseudolattices, del Pezzo surfaces, and Lefschetz fibrations*, Trans. Amer. Math. Soc. **373** (2020), 2071–2104.

[Hu16] D. Huybrechts, *Lectures on K3 surfaces*, Cambridge Univ. Press, 2016.

[Ku77] V. S. Kulikov, *Degenerations of K3 surfaces and Enriques surfaces*, Math. USSR-Izv. **11** (1977), 957–989.

[Kuz17] A. Kuznetsov, *Exceptional collections in surface-like categories*, Sb. Math. **208** (2017), 1368–1398.

[Ni80] V. V. Nikulin, *Integral symmetric bilinear forms and some of their applications*, Math. USSR-Izv. **14** (1980), 103–167.

[Se62] J.-P. Serre, *Formes bilinéaires symétriques entières à discriminant* ± 1, Sém. H. Cartan **14** (1961–62), exp. 14–15.

[SB83] N. I. Shepherd-Barron, *Extending polarizations on families of K3 surfaces*, in: The Birational Geometry of Degenerations, Progr. Math. **29**, Birkhäuser, 1983, 135–171.